

biomevae: Theory and Practice

Single-study, LOSO, strict LOSO, and the PhILR / TreeDTM model family

Generated from the repository implementation and documentation

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Deck scope

Practice

- Package inputs, outputs, CLI entry points.
- Single-study Snakemake pipeline.
- Non-strict LOSO pipeline.
- Strict LOSO pipeline.
- Model catalogue and evaluation artifacts.

Theory

- PhILR-VAE with full compositional geometry.
- TreeDTM-VAE with tree-local likelihoods.
- DIVA and PhyloDIVA.
- TAXI as taxonomy-protected conditional invariance.
- Why taxonomy regularization and domain invariance can conflict under label shift.

What biomevae trains

Model family	Main idea	Representative CLIs
Flat VAE baselines	MLP VAE on raw or log-transformed feature table	biomevae-train, biomevae-train-vanilla
Taxonomy-aware baselines	Taxonomy graph, tree prior, fusion, tree likelihoods	biomevae-train-graph, biomevae-train-tree-dtm, biomevae-train-philrvae
PhILR family	Aitchison-correct compositional coordinates from taxonomy SBP	biomevae-train-philrvae, biomevae-train-hyp-philrvae
DIVA wrappers	Latent split into domain, class, residual factors	biomevae-train-diva-*
PhyloDIVA	DIVA plus GRL, CORAL, phylogenetic smoothness	biomevae-train-phylodiva-*
TAXI	Protected taxonomy/class latent plus conditional residual invariance	biomevae-train-taxi-*

Canonical data contract

Per-study extracted inputs

- `sgb_table.tsv`: samples by taxa or SGB features.
- `phyla.tsv`: taxonomy rows aligned to feature clades.
- `sample_metadata.tsv`: labels, study name, covariates.

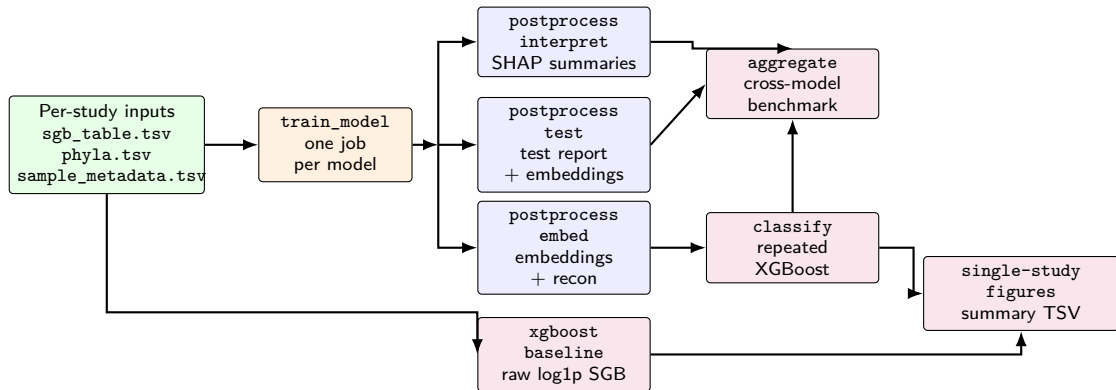
Important columns

- `disease`: default classification label.
- `study_name`: domain label in merged LOSO data.

Standard model outputs

- `model.pt`, `config.json`.
- `training_log.tsv`.
- `embeddings.tsv`; DIVA also writes `embeddings_z_d.tsv`, `embeddings_z_y.tsv`, `embeddings_z_x.tsv`.
- `recon.tsv`, `test_report.json`.
- Classification JSON and final summary TSVs.

Single-study Snakemake flow



Entry point: workflow/Snakefile. In single-study mode, cross-domain DIVA / PhyloDIVA / TAXI models are dropped because a single cohort gives a degenerate domain variable.

Single-study practice

Run

```
snakemake -s workflow/Snakefile \  
-configfile workflow/config/single_study.yaml \  
-cores 8
```

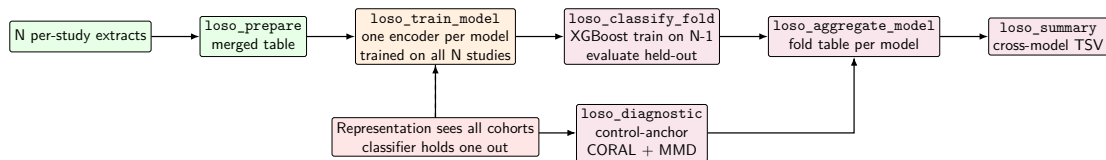
Key config

- data_root, output_root, study, label.
- extra_args: forwarded to every training CLI, defaulting to Optuna search.
- eval_seeds: default {42, 43, 44, 45, 46} for downstream evaluation.

Interpretation

- Best for within-study reconstruction, embedding quality, and phenotype prediction.
- Does not estimate cohort transfer because train and validation folds come from the same study.
- Cross-domain wrappers are intentionally excluded.

Non-strict LOSO flow



Entry point: `workflow/Snakefile.loso`. The encoder is shared across folds, so this is useful but not a fully strict deployment simulation.

Non-strict LOSO practice

Run

```
snakemake -s workflow/Snakefile.losos \
  -configfile workflow/config/losos_crc.yaml \
  -cores 16
```

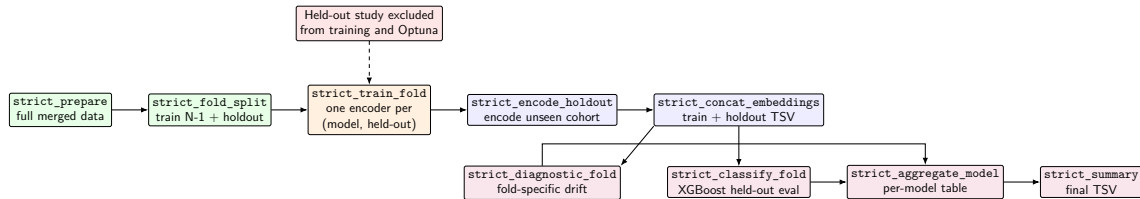
What it measures

- Downstream classifier transfer: train the classifier on all non-held-out studies and evaluate on the held-out study.
- Latent-space study drift: `losos_diagnostic` computes control-only CORAL and MMD.
- Model comparison under the same XGBoost classifier and five evaluation seeds.

Caveat

- The encoder and Optuna selection see the held-out cohort because the model is trained once on the merged dataset.
- This can be acceptable for exploratory representation analysis, but not for strict external validation.

Strict LOSO flow



Entry point: `workflow/Snakefile.losos_strict`. Cost is about N times non-strict training.

Strict LOSO practice

Run

```
snakemake -s workflow/Snakefile.losos_strict \  
-configfile workflow/config/losos_strict_crc.yaml \  
-cores 16
```

Why it is strict

- For each held-out cohort d , all model fitting uses only $D \neq d$.
- Optuna trials, validation splits, early stopping, and learned encoder weights never see the held-out cohort.
- Held-out samples are encoded only after training via `biomevae-losos_strict-encode`.

Cost

- Non-strict: one training run per model.
- Strict: N training runs per model, each usually with its own Optuna search.

LOSO model sweep

Group	Examples	Default classification slice
No domain adaptation	hyp-philrvae, tree-dtm-vae, beta-vae	full latent
DIVA	diva-tree-dtm-vae, diva-hyp-philr-nb, diva-beta-vae	z_y
PhyloDIVA	phylodiva-tree-dtm-vae, phylodiva-hyp-philr-nb, phylodiva-beta-vae	z_y
TAXI	taxi-tree-dtm-vae, taxi-hyp-philrvae	full predictive $z_\tau \cup z_\rho$
Tree baselines	xgb-baseline, xgb-coral	raw or CORAL-aligned SGB features

- DIVA / PhyloDIVA use z_y because it is the class-anchored factor.
- TAXI uses the protected plus residual predictive channels, not z_d .
- Slice behavior can be overridden in `loso_latent_slice`.

PhILR problem setting

For a sample with p taxa:

$$\mathbf{x} \in \mathbb{N}_0^p \quad \text{or} \quad \mathbf{x} \in \Delta^{p-1}, \quad \Delta^{p-1} = \{\mathbf{c} \in \mathbb{R}_{>0}^p : \sum_i c_i = 1\}.$$

Counts are closed to a composition with a pseudocount:

$$c_i = \frac{x_i + \alpha}{\sum_j (x_j + \alpha)}.$$

Modeling goal

- Separate composition from library depth.
- Use Euclidean latent-variable machinery in a coordinate system that is correct for compositional data.
- Build the coordinate system from the taxonomy tree, not arbitrary feature order.

Sequential binary partition balances

At an internal taxonomy split with descendant leaf groups R, S , cardinalities r, s , define a balance vector $\psi \in \mathbb{R}^p$:

$$\psi_i = \begin{cases} \sqrt{\frac{s}{r(r+s)}} & i \in R, \\ -\sqrt{\frac{r}{s(r+s)}} & i \in S, \\ 0 & \text{otherwise.} \end{cases}$$

- $\langle \psi, \mathbf{1} \rangle = 0$: balances live in the contrast subspace.
- $\|\psi\|_2 = 1$.
- Valid SBP contrasts are mutually orthogonal.
- Nodes with $K > 2$ children are linearized into $K - 1$ ordered left-versus-rest splits.

Stacking all balances gives

$$\Psi \in \mathbb{R}^{p \times (p-1)}, \quad \Psi^\top \Psi = I_{p-1}, \quad \text{span}(\Psi) = \mathbf{1}^\perp.$$

Forward and inverse PhILR

The Phylogenetic Isometric Log-Ratio transform is

$$\mathbf{y} = \text{ilr}(\mathbf{c}) = \log(\mathbf{c})\Psi \in \mathbb{R}^{p-1}.$$

The inverse used by the decoder is

$$\hat{\mathbf{c}} = \text{softmax}(\hat{\mathbf{y}}\Psi^\top) \in \Delta^{p-1}.$$

Metric consequence

$$\|\mathbf{y}^{(1)} - \mathbf{y}^{(2)}\|_2 = d_A(\mathbf{c}^{(1)}, \mathbf{c}^{(2)}),$$

where d_A is Aitchison distance. Therefore:

- A Gaussian in PhILR coordinates is a logistic-normal model on the simplex.
- MSE-like reconstruction in PhILR space is geometrically meaningful.
- No extra tree-consistency penalty is needed for closure.

PhILR-VAE encoder and decoder

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I_d)$$

$$q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag } \boldsymbol{\sigma}_\phi^2(\mathbf{x}))$$

$$\mathbf{x} \rightarrow \mathbf{c} \rightarrow \mathbf{y} = \log(\mathbf{c})\Psi \rightarrow h_\phi(\mathbf{y}) \rightarrow (\boldsymbol{\mu}_\phi, \log \boldsymbol{\sigma}_\phi^2)$$

$$\mathbf{z} = \boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I)$$

$$\mathbf{z} \rightarrow g_\theta(\mathbf{z}) = \hat{\mathbf{y}}$$

$$\hat{\mathbf{c}} = \text{softmax}(\hat{\mathbf{y}}\Psi^\top)$$

- Tree-aware likelihood variants also predict sibling-group concentration parameters.
- The same taxonomy backbone and node aggregation are shared with TreeDTM.

PhILR likelihoods

Likelihood	Distributional form	Use case
philr_gaussian	$\mathbf{y} \sim \mathcal{N}(\hat{\mathbf{y}}, \text{diag } \sigma_{\text{obs}}^2)$	relative abundance
multinomial	$\mathbf{x} \mid N, \hat{\mathbf{c}} \sim \text{Multinomial}(N, \hat{\mathbf{c}})$	fixed-depth counts
dirichlet_multinomial	$\alpha = \alpha_0 \hat{\mathbf{c}}$ then DM marginal	global overdispersion
dirichlet_tree_multinomial	product of DM terms over internal sibling groups	clade-specific count dispersion
dirichlet_tree	product of continuous Dirichlet densities over sibling fractions	relative abundance with tree likelihood

PhILR Gaussian reconstruction

For closed relative abundance input:

$$\mathbf{y} = \text{ilr}(\mathbf{c}), \quad \mathbf{y} \mid \mathbf{z} \sim \mathcal{N}(\hat{\mathbf{y}}_{\theta}(\mathbf{z}), \text{diag } \sigma_{\text{obs}}^2).$$

The negative log-likelihood is

$$-\log p_{\theta}(\mathbf{y} \mid \mathbf{z}) = \frac{1}{2} \sum_{k=1}^{p-1} \left[\frac{(y_k - \hat{y}_k)^2}{\sigma_{\text{obs},k}^2} + \log \sigma_{\text{obs},k}^2 \right] + \text{const.}$$

Interpretation

- This is the logistic-normal likelihood in the original simplex.
- The observation scales are learned per PhILR coordinate and clamped for stability.
- Library size does not enter because relative input has already been closed.

PhILR count likelihoods

Multinomial

$$\mathbf{x} \mid \hat{\mathbf{c}}, N \sim \text{Multinomial}(N, \hat{\mathbf{c}}), \quad -\log p = -\log \binom{N}{\mathbf{x}} - \sum_i x_i \log \hat{c}_i.$$

Dirichlet-multinomial

$$\boldsymbol{\alpha} = \alpha_0 \hat{\mathbf{c}}, \quad \alpha_0 > 0,$$
$$-\log p(\mathbf{x} \mid \boldsymbol{\alpha}, N) = -\log \binom{N}{\mathbf{x}} - \log \Gamma(\alpha_0) + \log \Gamma(N + \alpha_0) - \sum_i [\log \Gamma(x_i + \alpha_i) - \log \Gamma(\alpha_i)].$$

Tree variants

- PhILR encoder remains unchanged.
- Likelihood delegates to the TreeDTM node-aggregated sibling groups.

PhILR ELBO

For any supported likelihood:

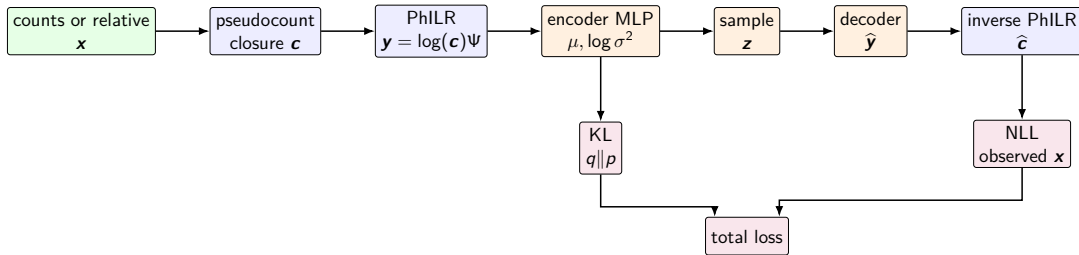
$$\mathcal{L}_{\text{PhILR}}^{(i)} = -\mathbb{E}_{q_\phi} \log p_\theta(\mathbf{x}^{(i)} \mid \mathbf{z}) + \beta_t \text{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}^{(i)}) \parallel \mathcal{N}(\mathbf{0}, I)) + \gamma \mathcal{R}(\theta_{\text{lik}}).$$

The Gaussian KL is

$$\text{KL} = \frac{1}{2} \sum_{j=1}^d (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2).$$

- β_t is the shared linear KL warm-up schedule.
- $\mathcal{R}$ regularizes likelihood-specific scales or concentrations.
- The objective is likelihood-agnostic except for the NLL term.

PhILR computation graph



Why PhILR is a new core model

- It replaces incoherent independent leaf NB heads for closed compositional data.
- It gives exact Aitchison geometry through an orthonormal taxonomy-derived basis.
- It keeps library size out of the latent except when the chosen count likelihood reintroduces N analytically.
- It can use five reconstruction likelihoods without changing the encoder geometry.
- It shares the same taxonomy graph machinery as TreeDTM, which keeps the training pipeline consistent across new backbones.

TreeDTM setup

Let $\mathcal{T}$ be a rooted taxonomy tree:

$$\mathcal{I} = \text{internal nodes}, \quad \mathcal{L} = \text{observed leaves}, \quad K_u = |\text{Ch}(u)|.$$

For each sample, leaf observations are aggregated to every node:

$$\mathbf{y} \in \mathbb{R}_{\geq 0}^N, \quad y_u = \sum_{v \in \text{Ch}(u)} y_v \quad (u \in \mathcal{I}).$$

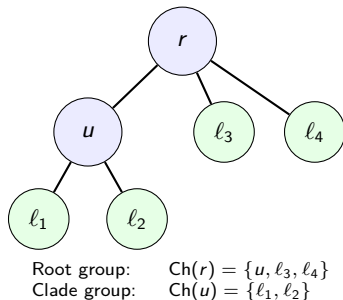
Each internal node u defines one sibling group

$$g = (u, \text{Ch}(u)), \quad K_g = K_u.$$

Core shift from old TreeNB

- Likelihood is placed at internal sibling splits, not independently at leaves.
- Dispersion is learned per clade / sibling group.
- Closure is enforced locally by multinomial or Dirichlet-family terms.

TreeDTM with a tree diagram



Observed node counts

$$y_u = y_{l_1} + y_{l_2}, \quad y_r = y_u + y_{l_3} + y_{l_4}.$$

Decoder probabilities

$$\mathbf{a} = h_\theta(\mathbf{z}), \quad \boldsymbol{\pi}_r = \text{softmax}(\mathbf{a}_{r \rightarrow u}, \mathbf{a}_{r \rightarrow l_3}, \mathbf{a}_{r \rightarrow l_4}),$$

$$\boldsymbol{\pi}_u = \text{softmax}(\mathbf{a}_{u \rightarrow l_1}, \mathbf{a}_{u \rightarrow l_2}).$$

Likelihood factorization

$$p(\mathbf{y} \mid \mathbf{z}) = p(\mathbf{x}_r \mid \boldsymbol{\pi}_r, \alpha_r^0) p(\mathbf{x}_u \mid \boldsymbol{\pi}_u, \alpha_u^0).$$

Each internal node contributes one local multinomial, Dirichlet-multinomial, or continuous Dirichlet term.

TreeDTM generative model

$$\mathbf{z}^{(i)} \sim \mathcal{N}(\mathbf{0}, I_d).$$

The decoder outputs edge or local split logits:

$$\mathbf{a}^{(i)} = h_{\theta}(\mathbf{z}^{(i)}) \in \mathbb{R}^E, \quad E = \sum_{u \in \mathcal{I}} K_u.$$

For every internal node u , a masked group softmax gives child probabilities:

$$\pi_{u \rightarrow v}^{(i)} = \frac{\exp(a_{u \rightarrow v}^{(i)})}{\sum_{v' \in \text{Ch}(u)} \exp(a_{u \rightarrow v'}^{(i)})}, \quad v \in \text{Ch}(u).$$

The same decoder probabilities support three likelihoods

- Tree-multinomial.
- Dirichlet-tree-multinomial.
- Continuous Dirichlet-tree.

Tree-multinomial likelihood

For internal node u , let child counts be

$$\mathbf{x}_u^{(i)} = (x_v^{(i)})_{v \in \text{Ch}(u)}, \quad n_u^{(i)} = \sum_{v \in \text{Ch}(u)} x_v^{(i)}.$$

The local likelihood is

$$p(\mathbf{x}_u^{(i)} \mid \boldsymbol{\pi}_{u,\cdot}^{(i)}) = \binom{n_u^{(i)}}{\mathbf{x}_u^{(i)}} \prod_{v \in \text{Ch}(u)} (\pi_{u \rightarrow v}^{(i)})^{x_{u \rightarrow v}^{(i)}}.$$

The joint log-likelihood is a product over internal nodes:

$$\log p(\mathbf{y}^{(i)} \mid \mathbf{z}^{(i)}) = \sum_{u \in \mathcal{I}} \log \text{Multinomial} \left(\mathbf{x}_u^{(i)}; n_u^{(i)}, \boldsymbol{\pi}_{u,\cdot}^{(i)} \right).$$

Dirichlet-tree-multinomial likelihood

The default TreeDTM count likelihood adds clade-local overdispersion. For sibling group g :

$$\alpha_g^{(i)} = \alpha_g^0 \pi_g^{(i)} + \epsilon, \quad \alpha_{g,0}^{(i)} = \sum_v \alpha_{g,v}^{(i)} \approx \alpha_g^0.$$

At internal node u corresponding to g :

$$p(\mathbf{x}_u^{(i)} \mid \alpha_g^{(i)}) = \binom{n_u^{(i)}}{\mathbf{x}_u^{(i)}} \frac{\Gamma(\alpha_{g,0}^{(i)})}{\Gamma(n_u^{(i)} + \alpha_{g,0}^{(i)})} \prod_{v \in \text{Ch}(u)} \frac{\Gamma(x_{u \rightarrow v}^{(i)} + \alpha_{g,v}^{(i)})}{\Gamma(\alpha_{g,v}^{(i)})}.$$

$$\log p(\mathbf{y}^{(i)} \mid \mathbf{z}^{(i)}) = \sum_{u \in \mathcal{I}} \log p(\mathbf{x}_u^{(i)} \mid \alpha_{g(u)}^{(i)}).$$

Continuous Dirichlet-tree likelihood

For relative-abundance data, each internal node contributes a Dirichlet density over observed child fractions:

$$\mathbf{q}_u^{(i)} = \text{normalise}(\mathbf{x}_u^{(i)} + \eta),$$
$$\log p(\mathbf{q}_u^{(i)} \mid \boldsymbol{\alpha}_g^{(i)}) = \log \Gamma(\alpha_{g,0}^{(i)}) - \sum_{v \in \text{Ch}(u)} \log \Gamma(\alpha_{g,v}^{(i)}) + \sum_{v \in \text{Ch}(u)} (\alpha_{g,v}^{(i)} - 1) \log q_{u \rightarrow v}^{(i)}.$$

- Internal nodes with zero parent mass are masked.
- η only keeps fractions inside the Dirichlet support.
- This gives a tree-native likelihood without pretending relative abundances are independent leaves.

TreeDTM encoder

The encoder consumes the node-aggregated tensor, then computes sibling-centered log-ratios:

$$\ell_{u \rightarrow v}^{(i)} = \log(y_v^{(i)} + \tau) - \frac{1}{K_u} \sum_{v' \in \text{Ch}(u)} \log(y_{v'}^{(i)} + \tau).$$

$$\ell^{(i)} = \text{concat}_{u \in \mathcal{I}, v \in \text{Ch}(u)} \ell_{u \rightarrow v}^{(i)}.$$

Then

$$(\mu_z^{(i)}, \log \sigma_z^{2(i)}) = g_\phi(\ell^{(i)}), \quad q_\phi(\mathbf{z} \mid \mathbf{y}) = \mathcal{N}(\mu_z, \text{diag } \sigma_z^2).$$

Why centered?

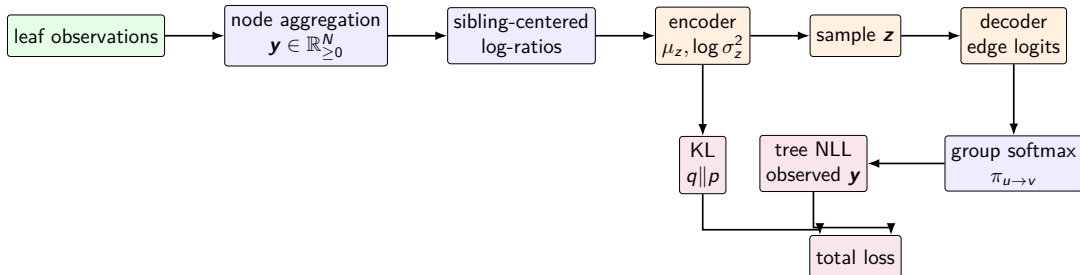
- Adding the same constant to all children of a node leaves ℓ unchanged.
- The encoder feature lives on the same local scale-invariant manifold as the likelihood.

TreeDTM training objective

$$\mathcal{L}_{\text{TreeDTM}}^{(i)} = -\log p_{\theta}(\mathbf{y}^{(i)} \mid \mathbf{z}^{(i)}) + \beta_t \text{KL}(q_{\phi}(\mathbf{z} \mid \mathbf{y}^{(i)}) \parallel \mathcal{N}(\mathbf{0}, I)) + \gamma \mathcal{R}(\boldsymbol{\alpha}^0).$$
$$\text{KL} = \frac{1}{2} \sum_j (\mu_{\mathbf{z},j}^2 + \sigma_{\mathbf{z},j}^2 - 1 - \log \sigma_{\mathbf{z},j}^2).$$

- Free-bits can floor KL per dimension.
- $\mathcal{R}(\boldsymbol{\alpha}^0)$ is the concentration L2 penalty.
- β_t uses the same warm-up resolution as the rest of the package.

TreeDTM computation graph



Why TreeDTM replaces TreeNB

Aspect	Deprecated TreeNB	TreeDTM
Likelihood location	independent leaves	internal sibling groups
Closure	decoder only	likelihood closes each sibling split
Dispersion	unrelated leaf parameters	per-clade concentration α_g^0
Encoder	leaf-only log counts	all-node sibling log-ratios
Data type	count-like only	counts or relative abundances
Statistical target	conflicts with compositional constraints	native tree compositional model

DIVA latent partition

DIVA factorizes the latent:

$$\mathbf{z} = [\mathbf{z}_d; \mathbf{z}_y; \mathbf{z}_x], \quad \mathbf{z}_d \in \mathbb{R}^{d_d}, \quad \mathbf{z}_y \in \mathbb{R}^{d_y}, \quad \mathbf{z}_x \in \mathbb{R}^{d_x}.$$

Conditional priors:

$$p(\mathbf{z}_d \mid d) = \mathcal{N}(\boldsymbol{\mu}_d(d), \text{diag } \boldsymbol{\sigma}_d^2(d)),$$

$$p(\mathbf{z}_y \mid y) = \mathcal{N}(\boldsymbol{\mu}_y(y), \text{diag } \boldsymbol{\sigma}_y^2(y)),$$

$$p(\mathbf{z}_x) = \mathcal{N}(\mathbf{0}, I).$$

Auxiliary classifiers:

$$q(d \mid \mathbf{z}_d) = \text{softmax } \psi_d(\mathbf{z}_d), \quad q(y \mid \mathbf{z}_y) = \text{softmax } \psi_y(\mathbf{z}_y).$$

DIVA objective

For a labelled sample $(\mathbf{x}, d, y)$:

$$\mathcal{L}_{\text{DIVA}}^{(i)} = \underbrace{-\log p_{\theta}(\mathbf{x}^{(i)} | \mathbf{z}^{(i)})}_{\text{backbone NLL}} + \beta_t (\text{KL}_d^{(i)} + \text{KL}_y^{(i)} + \text{KL}_x^{(i)}) + \alpha_d \text{CE}_d^{(i)} + \alpha_y \text{CE}_y^{(i)}.$$

$$\begin{aligned} \text{KL}_d &= \text{KL}(q_{\phi}(\mathbf{z}_d | \mathbf{x}) \| p(\mathbf{z}_d | d)), & \text{KL}_y &= \text{KL}(q_{\phi}(\mathbf{z}_y | \mathbf{x}) \| p(\mathbf{z}_y | y)), \\ \text{KL}_x &= \text{KL}(q_{\phi}(\mathbf{z}_x | \mathbf{x}) \| \mathcal{N}(\mathbf{0}, I)). \end{aligned}$$

For diagonal Gaussians:

$$\text{KL}(q \| p) = \frac{1}{2} \sum_j \left[\frac{\sigma_{q,j}^2 + (\mu_{q,j} - \mu_{p,j})^2}{\sigma_{p,j}^2} - 1 + \log \frac{\sigma_{p,j}^2}{\sigma_{q,j}^2} \right].$$

DIVA wrappers in biomevae

Wrapper	Backbone likelihood	Main use
DIVABetaVAE	Gaussian or MAE-like reconstruction on log-transformed counts	non-taxonomy domain-adaptation control
DIVAHyperbolicPhILRVAE	PhILR compositional likelihoods on Poincare latent geometry	compositional DIVA baseline
DIVATreeDTMVAE	tree-multinomial, Dirichlet-tree-multinomial, Dirichlet-tree	tree-likelihood DIVA baseline

- DIVA is likelihood-agnostic: the backbone supplies the NLL.
- Downstream LOSO classification normally reads z_y .
- Vanilla DIVA encourages z_d to encode domain, but does not explicitly scrub domain from z_y .

PhyloDIVA regularizers

PhyloDIVA adds three terms to DIVA:

$$\mathcal{L}_{\text{PhyloDIVA}} = \mathcal{L}_{\text{DIVA}} + \lambda_t \mathcal{L}_{\text{GRL}} + \mu \mathcal{L}_{\text{CORAL}} + \nu \mathcal{L}_{\text{BM}}.$$

1. Gradient-reversed study critic on z_y

$$\mathbf{z}_y \xrightarrow{\text{GRL}(\lambda_t)} \mathbf{z}'_y \xrightarrow{\text{MLP}} \hat{p}(d \mid \mathbf{z}_y), \quad \mathcal{L}_{\text{GRL}} = -\log \hat{p}(d \mid \mathbf{z}_y).$$

$$\lambda(t) = \lambda_{\max} \left(\frac{2}{1 + \exp(-\gamma t)} - 1 \right).$$

2. Deep CORAL on z_x

$$\mathcal{L}_{\text{CORAL}} = \frac{1}{|P|} \sum_{(k,k') \in P} \frac{1}{4p^2} \|\hat{\Sigma}_k - \hat{\Sigma}_{k'}\|_F^2.$$

PhyloDIVA tree smoothness

TreeDTM edge smoothness

$$\mathcal{L}_{\text{BM-edge}} = \frac{1}{|E^*|} \sum_{e \in E^*} \frac{(a_e - a_{\text{pa}(e)})^2}{\ell_e + \varepsilon}.$$

PhILR coordinate smoothness

$$\mathcal{L}_{\text{BM-coord}} = \frac{1}{|C^*|} \sum_{c \in C^*} \frac{(\hat{y}_c - \hat{y}_{\text{pa}(c)})^2}{\ell_c + \varepsilon}.$$

- With calibrated branch lengths, this is a Brownian-motion prior analogue.
- With uncalibrated taxonomies, it is better read as generic tree smoothness.
- The smoothness term is the taxonomy-preserving part of PhyloDIVA; the GRL and CORAL terms are the invariance-producing part.

DIVA versus PhyloDIVA diagnostics

Quantity	DIVA expectation	PhyloDIVA expectation
Domain accuracy from z_d Fresh domain classifier on z_y	high often high leakage	high near chance if adversary works
Covariance distance of z_x across studies Held-out class performance from z_y	nonzero may improve or leak	lower under CORAL should improve when study nuisance is separable

- These diagnostics are materialized by LOSO summaries and control-anchor reports.
- The next section explains why the last row can fail under label shift.

TAXI motivation

TAXI stands for Taxonomy-Anchored eXchangeable Invariance.

z_τ := protected taxonomy/class latent (implemented as DIVA z_y)

z_ρ := residual predictive latent (implemented as DIVA z_x)

z_d := study/domain reconstruction latent.

Instead of marginal invariance,

$$z_y \perp D,$$

TAXI enforces conditional residual invariance:

$$z_\rho \perp D \mid z_\tau, Y.$$

Principle

- Never pass z_τ through a GRL.
- Use z_τ and class context as detached conditioning information.
- Scrub only residual study signal from z_ρ .

TAXI conditional critic

The study critic receives

$$[\text{GRL}(z_\rho), \text{stopgrad}(z_\tau), \text{stopgrad}(y_{\text{ctx}})].$$

$$\mathcal{L}_{\text{cond_critic}} = -\log \hat{p}(d \mid \text{GRL}(z_\rho), z_\tau, y_{\text{ctx}}).$$

The class context is

$$y_{\text{ctx}} = \begin{cases} \text{onehot}(y), & \text{labelled sample,} \\ \text{softmax(class logits)}, & \text{unlabelled sample.} \end{cases}$$

- The critic learns to predict study.
- The reversed gradient reaches only z_ρ .
- The protected taxonomy/class channel z_τ remains predictive.

TAXI extra losses

$$\mathcal{L}_{\text{TAXI}} = \mathcal{L}_{\text{DIVA-backbone}} + \lambda_c \mathcal{L}_{\text{cond_critic}} + \lambda_{\text{coral}} \mathcal{L}_{\text{cond_CORAL}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{tree/philr}} + \lambda_{\text{orth}} \mathcal{L}_{\text{orth}} + \lambda_{\tau} \text{CE}_{\tau}.$$

Class-conditional CORAL on z_{ρ}

$$w_i^{(c,d)} = \mathbf{1}\{D_i = d\} p_i(Y = c), \quad \hat{\Sigma}_{c,d} = \text{Cov}_{w^{(c,d)}}(z_{\rho}),$$

$$\mathcal{L}_{\text{cond_CORAL}} = \text{mean}_{c,d < d'} \frac{\|\hat{\Sigma}_{c,d} - \hat{\Sigma}_{c,d'}\|_F^2}{4p^2}.$$

Orthogonality

$$\mathcal{L}_{\text{orth}} = \text{mean}_{c,d} \|\text{Cov}_{w^{(c,d)}}(z_{\tau}, z_{\rho})\|_F^2.$$

TAXI wrappers

Wrapper	Backbone	Main difference from Phylo-DIVA
TAXIDIVATreeDTMVAE	DIVA TreeDTM	conditional critic on z_ρ , tree contrast smoothness, class-conditional CORAL
TAXIHyperbolicPhILRVAE	DIVA Hyperbolic PhILR	conditional critic on z_ρ , PhILR coordinate smoothness, class-conditional CORAL

- The classifier reads $[z_\tau, z_\rho]$.
- z_d remains the domain reconstruction latent.
- CLI aliases accept `-latent-tau/-latent-rho` and also `-latent-y/-latent-x`.
- Default LOSO slice is `full` for TAXI so the saved embedding contains the predictive concatenation.

The conflict

Two desirable biases can point in opposite directions.

Taxonomy-aware learning

- Disease signal is smooth on a taxonomy tree.
- PhILR and TreeDTM preserve tree-compositional structure.
- Smoothness reduces variance and improves finite-sample transfer.

The hard case is label shift:

$$\mathcal{L}(Y \mid D = d) \neq \mathcal{L}(Y \mid D = d').$$

Domain invariance

- A held-out study should not be identifiable from the predictive latent.
- GRL and CORAL reduce residual study dependence.
- If study identity and disease prevalence are linked, invariance can erase predictive taxonomic signal.

Formal setup

Random variables:

X = taxa table, D = study, Y = phenotype, $A(X)$ = taxonomic state carrying signal.

For two studies d, d' , define

$$\Delta_Y = \text{TV}(\mathcal{L}(Y \mid D = d), \mathcal{L}(Y \mid D = d')),$$

$$\Delta_A = \text{TV}(\mathcal{L}(A \mid D = d), \mathcal{L}(A \mid D = d')).$$

Residual study dependence of an invariant latent:

$$\delta = \text{TV}(\mathcal{L}(Z_{\text{inv}} \mid D = d), \mathcal{L}(Z_{\text{inv}} \mid D = d')) .$$

- PhyloDIVA tries to drive δ down.
- LOSO diagnostics estimate the practical size of taxonomy and latent drift.

Taxonomy-aware upper bound

Assume a tree-smooth target f^* with approximation error $\varepsilon_{\mathcal{T}}^{\text{approx}}$:

$$\Pr\left(Y \neq \arg \max_c f_{A(X),c}^*\right) \leq \varepsilon_{\mathcal{T}}^{\text{approx}}, \quad \Omega_{\mathcal{T}}(f^*) \leq B^2.$$

For a taxonomy-aware predictor,

$$R_d(\hat{Y}_{\mathcal{T}}) \leq \varepsilon_{\mathcal{T}}^{\text{approx}} + c_1 \sqrt{\frac{d_{\mathcal{T}}(\lambda)}{n}} + c_2 \sqrt{\frac{\log(1/\eta)}{n}} + \lambda B^2$$

where

$$d_{\mathcal{T}}(\lambda) = \text{tr}(L_{\mathcal{T}}(L_{\mathcal{T}} + \lambda I)^{-1}).$$

Informative taxonomy means $d_{\mathcal{T}}(\lambda) \ll p - 1$, so the tree prior reduces variance.

Invariant lower bound

For any classifier using an invariant latent with

$$\text{TV}(\mathcal{L}(Z_{\text{inv}} \mid D = d), \mathcal{L}(Z_{\text{inv}} \mid D = d')) \leq \delta,$$

the balanced two-study risk obeys

$$\frac{1}{2} \left[R_d(\hat{Y}_{\text{inv}}) + R_{d'}(\hat{Y}_{\text{inv}}) \right] \geq \frac{1}{2}(\Delta_Y - \delta)_+.$$

Reading

- If the adversary succeeds, $\delta \rightarrow 0$.
- Under label shift, the lower bound approaches $\Delta_Y/2$.
- Stronger invariance can tighten the lower bound against the model.

Crossover condition

Taxonomy-aware prediction wins when

$$\frac{1}{2}(\Delta_Y - \delta)_+ > \varepsilon_{\mathcal{T}}^{\text{approx}} + c_1 \sqrt{\frac{d_{\mathcal{T}}(\lambda)}{n}} + c_2 \sqrt{\frac{\log(1/\eta)}{n}} + \lambda B^2.$$

Operational meaning

- If $\Delta_Y \approx \Delta_A$, the disease label shift is carried by taxonomic composition.
- A marginal GRL on z_Y cannot distinguish “study as nuisance” from “study as taxonomic disease signal”.
- PhILR / TreeDTM without aggressive invariance may outperform PhyloDIVA.
- TAXI is designed for this regime: protect z_{τ} , align only z_{ρ} conditionally.

Binary worked example

Two studies, binary phenotype, binary taxonomic state:

$$Y = A, \quad \Pr(A = 1 \mid D = 0) = 0.1, \quad \Pr(A = 1 \mid D = 1) = 0.9.$$

Then

$$\Delta_Y = \Delta_A = 0.8.$$

Taxonomy-aware model

- PhILR or TreeDTM can recover the marker clade.
- Risk is near $\varepsilon_{\mathcal{T}}^{\text{approx}}$, up to finite-sample error.

Strict marginal invariance

- If $\delta = 0$, the invariant lower bound is at least 0.4.
- In the symmetric example, any domain-independent output has balanced risk 0.5.

Practical knob interpretation

Knob	Helps taxonomy bound	Affects invariance bound
-smooth-weight	lowers effective tree complexity when tuned well	no direct effect on δ
-grl-lambda	no direct effect	reduces δ , which can hurt under label shift
-coral-weight	no direct effect	reduces second-order study drift
-alpha-y	sharpens z_y as class latent	may increase study dependence if labels are study-shifted
TAXI -lambda-cond-*	preserves z_τ signal	aligns only residual z_ρ conditionally

- Use control-anchor diagnostics to decide whether study drift is nuisance or taxonomic signal.
- Compare strict and non-strict LOSO to quantify representation-side leakage.

Choosing a pipeline

Goal	Use	Caveat
Within-study benchmarking	Single-study workflow/Snakefile	no external cohort transfer
Fast cross-cohort screening	Non-strict Snakefile.los	encoder sees all cohorts
Deployment-like external validation	Strict Snakefile.los_strict	N -times more model training
Diagnose study drift	LOSO diagnostics	compare within class, especially controls
Attribute domain adaptation gains	compare beta, DIVA-beta, XGB, XGB-CORAL rows	same labels and seeds required

Choosing a model

- Start with `tree-dtm-vae` and `philrvae/hyp-philrvae` when the data are compositional and taxonomy is reliable.
- Use DIVA when domain effects are strong and plausibly separable from phenotype signal.
- Use PhyloDIVA when you want DIVA plus explicit latent study scrubbing, CORAL residual alignment, and tree smoothness.
- Use TAXI when the phenotype signal is likely taxonomic and study-shifted, so marginal z_y invariance may over-scrub.
- Always compare against `xgb-baseline`; a VAE row that does not beat raw-feature XGBoost is not earning its complexity.

Artifacts to inspect

Artifact	What to read
training_log.tsv	convergence, KL warm-up, auxiliary and adaptation losses
config.json	selected hyperparameters, latent dimensions, likelihood
embeddings*.tsv	full latent and DIVA/TAXI slices
classification_results.json	per-seed XGBoost metrics and fold sizes
loso_summary.tsv	cross-model held-out performance table
control_anchor_summary.json	control-only MMD and CORAL drift diagnostics
optuna_best_params.json	best trial configuration after HPO

Implementation map

Concept	Code
PhILR basis and transform	<code>src/biomevae/models/philrvae.py</code> , <code>taxonomy_tree.py</code>
TreeDTM likelihood and topology	<code>src/biomevae/models/tree_dtm_vae.py</code>
DIVA core	<code>src/biomevae/models/diva.py</code>
PhyloDIVA adaptation losses	<code>src/biomevae/models/phylo_da.py</code> , <code>phylo_cov.py</code>
TAXI conditional invariance	<code>src/biomevae/models/taxi_treedtmvae.py</code> , <code>taxi_hyp_philrvae.py</code>
LOSO preparation and classification	<code>src/biomevae/loso.py</code> , <code>src/biomevae/cli/loso_classify.py</code>
Snakemake rules	<code>workflow/rules/*.smk</code>

Takeaways

- PhILR-VAE is the Aitchison-correct compositional backbone: taxonomy defines an orthonormal ILR basis, and likelihoods respect either the simplex or fixed-depth count table.
- TreeDTM-VAE is the tree-native count/relative likelihood backbone: reconstruction happens through local sibling distributions and clade-level overdispersion.
- DIVA partitions representation by domain, label, and residual factors; PhyloDIVA strengthens invariance and smoothness.
- TAXI protects the taxonomy/class channel and aligns only the residual predictive channel conditionally.
- Strict LOSO is the decisive external-validation pipeline; non-strict LOSO is useful for screening but has representation-side leakage.
- Under label shift, taxonomy regularization and domain invariance can be mathematically opposed. Diagnostics should guide which bias to trust.

References in the repository

- docs/philrvae_theory.md
- docs/tree_dtm_vae_theory.md
- docs/diva_theory.md
- docs/phylodiva_theory.md
- docs/taxonomy_vs_domain_invariance_theorem.md
- workflow/README.md and workflow/README_loso.md
- workflow/Snakefile, workflow/Snakefile.loso, workflow/Snakefile.loso_strict

External methods named in the docs: DIVA (Ilse et al. 2020), DANN/GRL (Ganin and Lempitsky 2015), Deep CORAL (Sun and Saenko 2016), Brownian-motion tree priors (Felsenstein 1985), and domain-adaptation lower bounds.